Boolean Algebra

Axioms and Laws of Boolean Algebra

Axiom 1 :	$0 \cdot 0 = 0$	Axiom 6 :	$0 + 1 = 1$
Axiom 2 :	$0 \cdot 1 = 0$	Axiom 7 :	$1 + 0 = 1$
Axiom 3 :	$1 \cdot 0 = 0$	Axiom 8 :	$1 + 1 = 1$
Axiom 4 :	$1 \cdot 1 = 1$	Axiom 9 :	$\overline{1} = 0$
Axiom 5 :	$0 + 0 = 0$	Axiom 10 :	$\overline{0} = 1$

Complementation Laws

$$\overline{0} = 1$$

$$\overline{1} = 0$$

If $A = 0$, then $\overline{A} = 1$

If $A = 1$, then $\overline{A} = 0$

$$\overline{\overline{A}} = A$$

AND Laws

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A \cdot A = A$$

$$A \cdot \overline{A} = 0$$

OR Laws

$$A + 0 = A$$

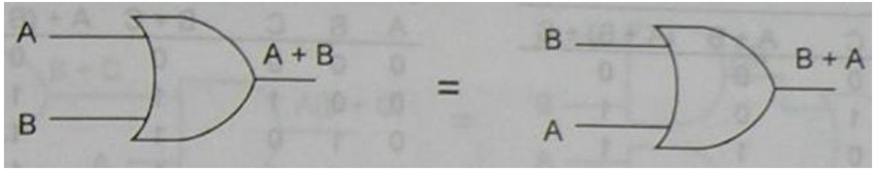
$$A + 1 = 1$$

$$A + A = A$$

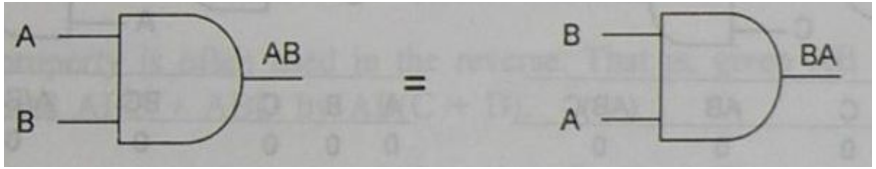
$$A + \overline{A} = 1$$

Commutative Laws

$$A + B = B + A$$



$$A \cdot B = B \cdot A$$



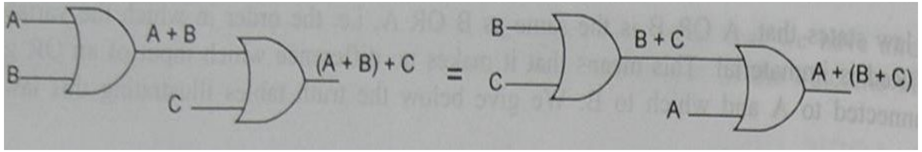
- Except NOT gate, all gates follow commutative law
- Can be extended to any number of variables

$$A + B + C = B + C + A = C + A + B = B + A + C$$

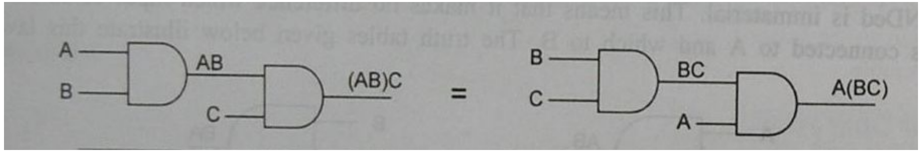
$$A \cdot B \cdot C = B \cdot C \cdot A = C \cdot A \cdot B = B \cdot A \cdot C$$

Associative Laws

$$(A + B) + C = A + (B + C)$$



$$(A \cdot B)C = A(B \cdot C)$$



- Can be extended to any number of variables

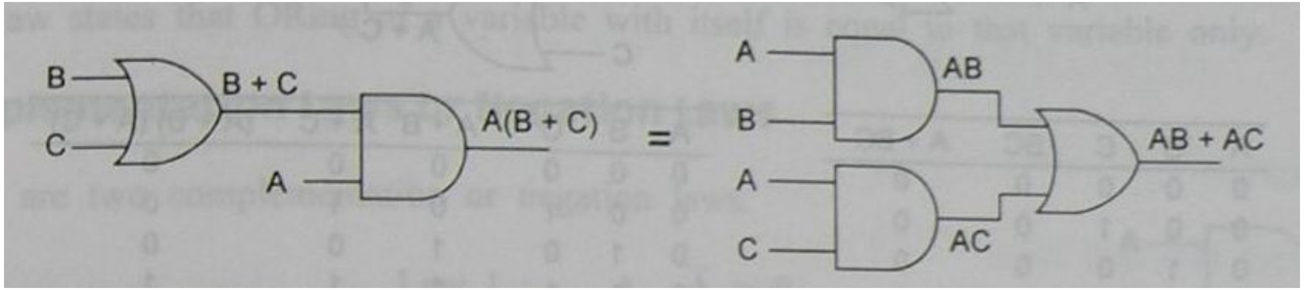
$$A + (B + C + D) = (A + B + C) + D = (A + B) + (C + D).$$

$$A(BCD) = (ABC)D = (AB)\ (CD)$$

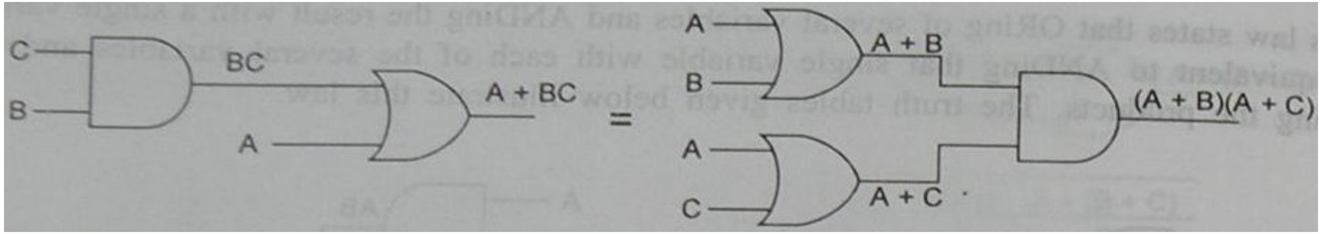
- NAND and NOR gates do not follow associative law
- XNOR follows associative law for even number of variables not for odd number of variables

Distributive Laws

$$A(B + C) = AB + AC$$

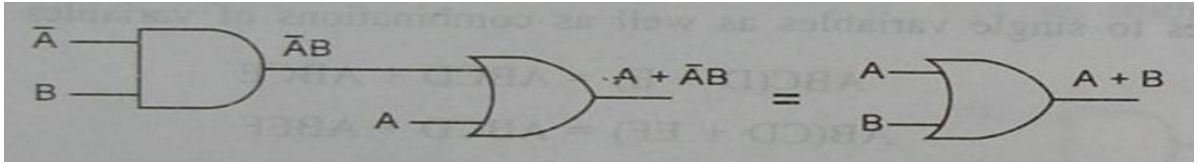


$$A + BC = (A + B)(A + C)$$



$$\begin{aligned}
RHS &= (A + B)(A + C) \\
&= AA + AC + BA + BC \\
&= A + AC + AB + BC \\
&= A(1 + C + B) + BC \\
&= A \cdot 1 + BC \qquad (\because 1 + C + B = 1 + B = 1) \\
&= A + BC \\
&= LHS
\end{aligned}$$

$$A + \overline{A}B = A + B$$



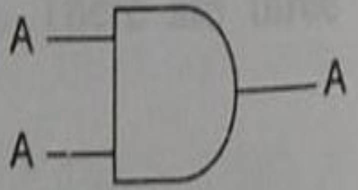
$$
\begin{aligned}
A + \overline{A}B &= (A + \overline{A})\,(A + B) \\
&= 1 \cdot (A + B) \\
&= A + B
\end{aligned}
$$

Idempotence Laws

$$A \cdot A = A$$

If $A = 0$, then $A \cdot A = 0 \cdot 0 = 0 = A$

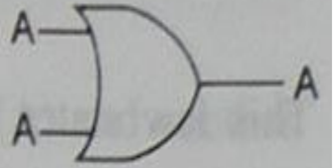
If $A = 1$, then $A \cdot A = 1 \cdot 1 = 1 = A$



$$A + A = A$$

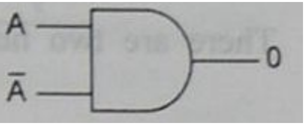
If $A = 0$, then $A + A = 0 + 0 = 0 = A$

If $A = 1$, then $A + A = 1 + 1 = 1 = A$



Complementation or Negation Laws

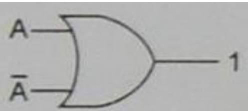
$A . \overline{A} = 0$



If $A = 0$, then $\overline{A} = 1$ and $A \cdot \overline{A} = 0 \cdot 1 = 0$

If $A = 1$, then $\overline{A} = 0$ and $A \cdot \overline{A} = 1 \cdot 0 = 0$

$A + \overline{A} = 1$



If $A = 0$, then $\overline{A} = 1$ and $A + \overline{A} = 0 + 1 = 1$

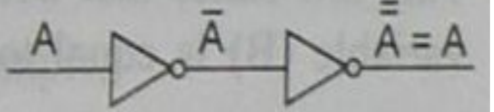
If $A = 1$, then $\overline{A} = 0$ and $A + \overline{A} = 1 + 0 = 1$

Double Negation Laws

$$\overline{\overline{A}} = A$$

If $A = 0$, then $\overline{\overline{A}} = \overline{\overline{0}} = \overline{1} = 0 = A$

If $A = 1$, then $\overline{\overline{A}} = \overline{\overline{1}} = \overline{0} = 1 = A$



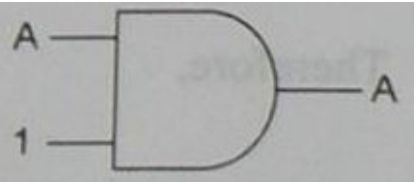
- Any odd number of inversions = Single inversion
- Any even number of inversions = No inversion at all

Identity Laws

$$A \cdot 1 = A$$

if $A = 0$, then $A \cdot 1 = 0 \cdot 1 = 0 = A$

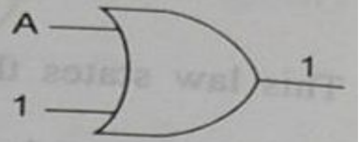
if $A = 1$, then $A \cdot 1 = 1 \cdot 1 = 1 = A$



$$A + 1 = 1$$

if $A = 0$, then $A + 1 = 0 + 1 = 1$

if $A = 1$, then $A + 1 = 1 + 1 = 1$

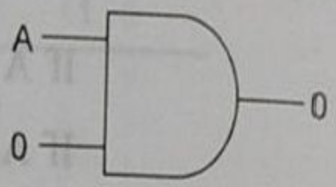


Null Laws

$$A \cdot 0 = 0$$

if $A = 0$, then $A \cdot 0 = 0 \cdot 0 = 0$

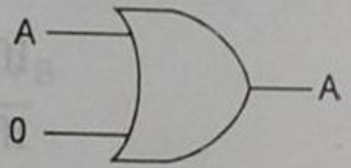
if $A = 1$, then $A \cdot 0 = 1 \cdot 0 = 0$



$$A + 0 = A$$

if $A = 0$, then $A + 0 = 0 + 0 = 0 = A$

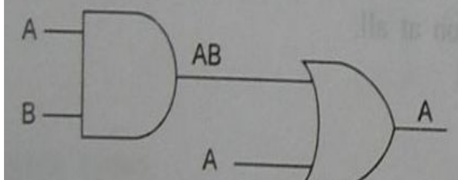
if $A = 1$, then $A + 0 = 1 + 0 = 1 = A$



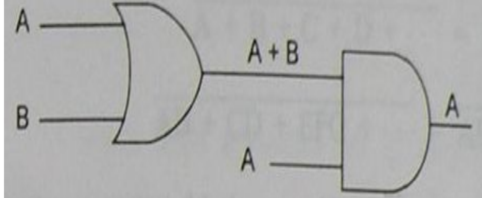
Absorption Laws

$$A + A \cdot B = A$$

$$A(A + B) = A$$



A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1



A	B	A + B	A(A + B)
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

$$A + A \cdot B = A(1 + B) = A \cdot 1 = A$$

$$A + A \cdot \text{Any term} = A$$

$$A(A + B) = A \cdot A + A \cdot B = A + AB = A(1 + B) = A \,.\, 1 = A$$

$$A(A + \text{Any term}) = A$$

Consensus Theorem

$$AB + \overline{A}C + BC = AB + \overline{A}C$$

$$\begin{aligned}
LHS &= AB + \overline{A}C + BC \\
&= AB + \overline{A}C + BC(A + \overline{A}) \\
&= AB + \overline{A}C + BCA + BC\overline{A} \\
&= AB(1 + C) + \overline{A}C(1 + B) \\
&= AB(1) + \overline{A}C(1) \\
&= AB + \overline{A}C \\
&= RHS
\end{aligned}$$

$$(A + B)(\overline{A} + C)(B + C) = (A + B)(\overline{A} + C)$$

$$\begin{aligned}
LHS &= (A + B)(\overline{A} + C)(B + C) = (A\overline{A} + AC + B\overline{A} + BC)(B + C) \\
&= (AC + BC + \overline{A}B)(B + C) \\
&= ABC + BC + \overline{A}B + BC + AC + \overline{A}BC = AC + BC + \overline{A}B
\end{aligned}$$

$$\begin{aligned}
RHS &= (A + B)(\overline{A} + C) \\
&= A\overline{A} + AC + BC + \overline{A}B \\
&= AC + BC + \overline{A}B = LHS
\end{aligned}$$

- Can be extended to any number of variables

$$AB + \overline{A}C + BCD = AB + \overline{A}C$$

$$LHS = AB + \overline{A}C + BCD = AB + \overline{A}C + BC + BCD = AB + \overline{A}C + BC = AB + \overline{A}C = RHS$$

$$(A + B)(\overline{A} + C)(B + C + D) = (A + B)(\overline{A} + C)$$

$$\begin{aligned}
LHS &= (A + B)(\overline{A} + C)(B + C)(B + C + D) = (A + B)(\overline{A} + C)(B + C) \\
&= (A + B)(\overline{A} + C)
\end{aligned}$$

Conditions

1. 3 Variables
2. Each variable used twice as pair
3. Only 1 variable is complemented one time

$AB + A'C + BC = AB + BC$

Transposition Theorem

$$AB + \overline{A}C = (A + C)(\overline{A} + B)$$

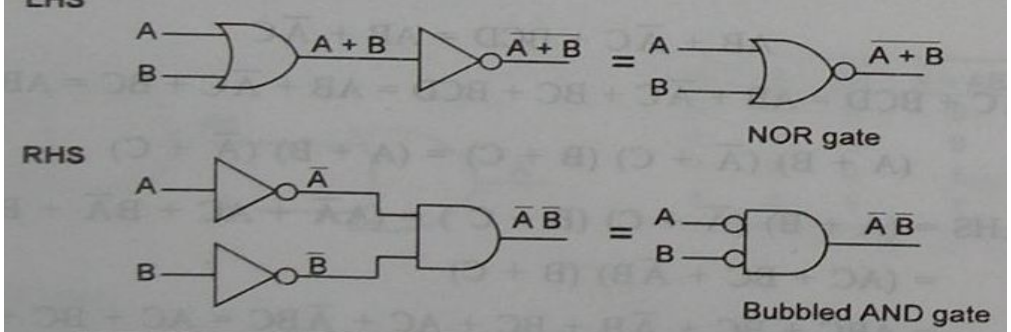
$$\begin{aligned}
\text{RHS} &= (A + C)(\overline{A} + B) \\
&= A\overline{A} + C\overline{A} + AB + CB \\
&= 0 + \overline{A}C + AB + BC \\
&= \overline{A}C + AB + BC(A + \overline{A}) \\
&= AB + ABC + \overline{A}C + \overline{A}BC \\
&= AB + \overline{A}C \\
&= \text{LHS}
\end{aligned}$$

(A+C) (A'+B) = AB + A'C

AB + A'C = (A+C) (A'+B)

De Morgan’s Theorem

$$\overline{A + B} = \overline{A}\,\overline{B}$$



A	B	A + B	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

=

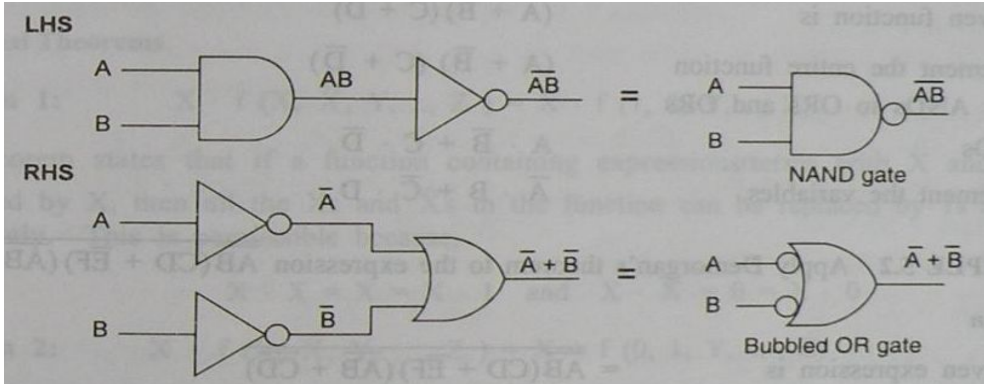
A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

- Can be extended to any number of variables or combination of variables

$$\overline{A + B + C + D + \cdots} = \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} \ldots$$

$$\overline{AB + CD + EFG + \cdots} = \overline{AB}\ \overline{CD}\ \overline{EFG} \ldots$$

$$\overline{AB} = \overline{A} + \overline{B}$$



A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

=

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

- Can be extended to any number of variables or combination of variables

$$\overline{ABCD\ldots} = \overline{A} + \overline{B} + \overline{C} + \overline{D} + \ldots$$

$$\overline{(AB)(CD)(EFG)\ldots} = \overline{AB} + \overline{CD} + \overline{(EFG)} + \ldots$$

Demorganization

1. Complement the entire given function
2. Change all the ANDs to ORs and all the ORs to ANDs
3. Complement each of the individual variables

Dual

- To convert positive logic level into negative logic level and vice versa
- Only logic is changed, voltage remain unchanged

Positive logic	**Negative logic**
Logic ‘0’: 0V	Logic ‘0’: 5V
Logic ‘1’: 5V	Logic ‘1’: 0V
Logic ‘0’: -5V	Logic ‘0’: 0V
Logic ‘1’: 0V	Logic ‘1’: -5V

+ve logic AND

A	**B**	**Y**
0	0	0
0	1	0
1	0	0
1	1	1

Logic ‘0’ = 0V
Logic ‘1’ = 5V

-ve logic OR

A	**B**	**Y**
1	1	1
1	0	1
0	1	1
0	0	0

Logic ‘0’ = 5V
Logic ‘1’ = 0V

Rules

1. AND ⟷ OR
   ‘.’ ⟷ ‘+’
2. 1 ⟷ 0
3. Keep variables as it is

Duals

AND ⟷ OR
NAND ⟷ NOR
XOR ⟷ XNOR
NOT ⟷ NOT
BUFFER ⟷ BUFFER

Given Expression	*Dual*
1. $\overline{0} = 1$	$\overline{1} = 0$
2. $0 \cdot 1 = 0$	$1 + 0 = 1$
3. $0 \cdot 0 = 0$	$1 + 1 = 1$
4. $1 \cdot 1 = 1$	$0 + 0 = 0$
5. $A \cdot 0 = 0$	$A + 1 = 1$
6. $A \cdot 1 = A$	$A + 0 = A$
7. $A \cdot A = A$	$A + A = A$
8. $A \cdot \overline{A} = 0$	$A + \overline{A} = 1$
9. $A \cdot B = B \cdot A$	$A + B = B + A$
10. $A \cdot (B \cdot C) = (A \cdot B) \cdot C$	$A + (B + C) = (A + B) + C$
11. $A \cdot (B + C) = AB + AC$	$A + BC = (A + B)(A + C)$
12. $A(A + B) = A$	$A + AB = A$
13. $A \cdot (A \cdot B) = A \cdot B$	$A + A + B = A + B$
14. $\overline{AB} = \overline{A} + \overline{B}$	$\overline{A + B} = \overline{A}\ \overline{B}$
15. $(A + B)(\overline{A} + C)(B + C) = A + B(\overline{A} + C)$	$AB + \overline{A}C + BC = AB + \overline{A}C$

Complement

A	**B**	**Y=AB**	**Y'=A'+B'**
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Example: Y = ABC + A'BC +ABC'
Y' = (A'+B'+C').(A+B'+C').(A'+B'+c)

Rules

1. AND $\longleftrightarrow$ OR
   '.' $\longleftrightarrow$ '+'
2. 1 $\longleftrightarrow$ 0 (Only in O/P)
3. Complement each variable

Operator Precedence

1. Parentheses
2. NOT
3. AND
4. OR

EXAMPLE 5.2 Apply Demorgan's theorem to the expression $\overline{\overline{AB}(CD + \overline{E}F)(\overline{AB} + \overline{CD})}$

EXAMPLE 5.3 Reduce the expression $\overline{\overline{AB} + \overline{A} + AB}$

EXAMPLE 5.2 Apply Demorgan's theorem to the expression $\overline{\overline{AB}(CD + \overline{E}F)(\overline{AB} + \overline{CD})}$

Solution

The given expression is

$$= \overline{\overline{AB}(CD + \overline{E}F)(\overline{AB} + \overline{CD})}$$

$$= \overline{\overline{AB}} + \overline{CD + \overline{\overline{E}F}} + (\overline{\overline{AB} + \overline{CD}})$$

$$= AB + (\overline{CD} \cdot \overline{\overline{E}F}) + (\overline{\overline{AB}} \cdot \overline{\overline{CD}})$$

$$= AB + (\overline{C} + \overline{D})(E + \overline{F}) + ABCD$$

A second method of performing demorganization is "Break the line, change the sign". For example, if we wish to demorganize the expression $\overline{AB} + \overline{CDE}$, we can break the line between A and B, and the line between C and D, and that between D and E and change the sign from ANDing to ORing. This yields $\overline{A} + \overline{B} + \overline{C} + \overline{D} + \overline{E}$.

EXAMPLE 5.3 Reduce the expression $\overline{\overline{AB} + \overline{A} + AB}$

Solution

The given expression is $\overline{\overline{AB} + \overline{A} + AB}$

Break the upper line between $\overline{AB}$ and $\overline{A}$, and that between $\overline{A}$ and AB, and change the signs between $\overline{AB}$, $\overline{A}$, AB.

$$= \overline{\overline{AB}} \cdot \overline{\overline{A}} \cdot \overline{AB}$$

$$= AB \cdot A \cdot \overline{AB}$$

Simplify

$$= AB \cdot \overline{AB}$$

$$= 0$$

Alternatively:

Break the lower line between A and B and change the sign between them.

$$= \overline{\overline{A} + \overline{B} + {}}\,\overline{\overline{A} + AB}$$

Simplify

$$= \overline{\overline{A} + \overline{B} + AB}$$

Break the line between $\overline{A}$ and $\overline{B}$, and that between $\overline{B}$ and AB, and change the sign between them.

$$= \overline{\overline{A}} \cdot \overline{\overline{B}} \cdot \overline{AB} = AB \cdot \overline{AB} = 0$$

1. $(X+Y+Z)(X+Y'+Z)(X+Y+Z')$
2. Let $f(A,B) = A'+B$ then find $f(f(X+Y,Y),Z)$

1. (X+Y+Z) (X+Y’+Z) (X+Y+Z’)

Sol: (X+Y)(X+Y’+Z)

= X+YY’+YZ

=X+YZ

2. Let f(A,B) = A’+B then find f(f(X+Y,Y),Z)

Sol: f(A+B) = A’+B

f(X+Y,Y) = (X+Y)’+Y = X’Y’+Y = X’+Y

f(f(X+Y,Y),Z) = f(X’+Y,Z) = (X’+Y)’+Z

=X.Y’+Z